

Scenario Evaluation Worksheet

Instructions: Use this worksheet to document your AI product concept analysis.

Part I: Analyze and document opportunities

Product Concept Opportunities
Frictionless shopping experience Customers can shop without waiting in checkout lines, increasing convenience and satisfaction.
Real-time cart updates The app shows item costs instantly, improving shopping accuracy and reducing billing disputes.
Higher store throughput Checkout-free systems reduce bottlenecks, enabling more customers to shop simultaneously.
Improved operational efficiency Automation reduces cashier labor costs and allows staff to focus on customer service and store operations.
Data-driven insights for the company Customer behavior, popular items, dwell time, and traffic patterns can be used to optimize inventory and store layout.
Competitive differentiation Offers a modern, AI-powered experience that can attract tech-savvy shoppers and provide a unique brand advantage.
Potential for scalability across many locations Once the system is built, it can be replicated in multiple stores with lower incremental costs.

Part II: Analyze and document challenges

Product Concept Challenges
High cost of sensors, hardware, and infrastructure Installing cameras, sensors, edge devices, and cloud infrastructure requires a major investment.
Complexity of real-time object recognition & tracking The system must accurately track customers, carts, and items under varying conditions (lighting, movement, crowding).
Data privacy and security concerns Continuous monitoring raises GDPR/CCPA considerations, requiring secure handling of customer data.
Accuracy issues during peak hours Dense crowds can cause system errors such as misidentification of items or incorrect cart updates.

Integration with payment systems and fraud prevention Real-time payment processing must be secure, reliable, and compliant with financial regulations.
Customer trust and adoption barriers Some shoppers may distrust automated billing or feel uncomfortable with camera-based tracking.
Ongoing maintenance and system updates Cameras, sensors, and AI models will require continuous calibration and retraining.

Part III Identify the AI product management team

AI Product Team Role	Team Member Responsibilities
AI Product Manager	Defines vision, requirements, prioritization, user stories, success metrics; aligns stakeholders; manages roadmap; ensures product solves customer and business problems.
Data Scientist / ML Engineer	Builds, trains, tests, and optimizes object detection, tracking, and prediction models; evaluates accuracy; handles model drift and retraining.
Computer Vision Engineer	Designs and implements vision models for item detection, customer tracking, and sensor fusion; ensures real-time performance and reliability.
Data Engineer	Manages data pipelines, sensor integrations, storage, and processing infrastructure; ensures availability and quality of data for AI models.
Software Engineer / Mobile Developer	Builds the customer-facing app, integrates with APIs, handles authentication, updates cart UI in real time, and manages scalability.
UX/UI Designer	Designs simple onboarding, real-time cart UI, intuitive experience, and customer feedback flow.
Security & Compliance Specialist	Ensures privacy, compliance with data regulations, secure handling of payment and identity data.